

LESSON 7

BOOK
CHAPTER 25

CAPACITANCE

Capacitors in series combination:

Potential difference of each actual capacitor:

$$V_1 = \frac{q}{C_1} \quad V_2 = \frac{q}{C_2} \quad \text{And} \quad V_3 = \frac{q}{C_3}$$

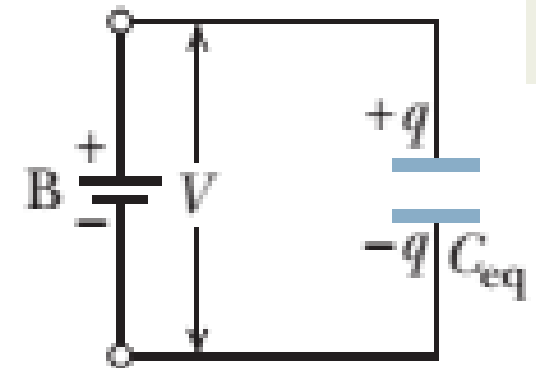
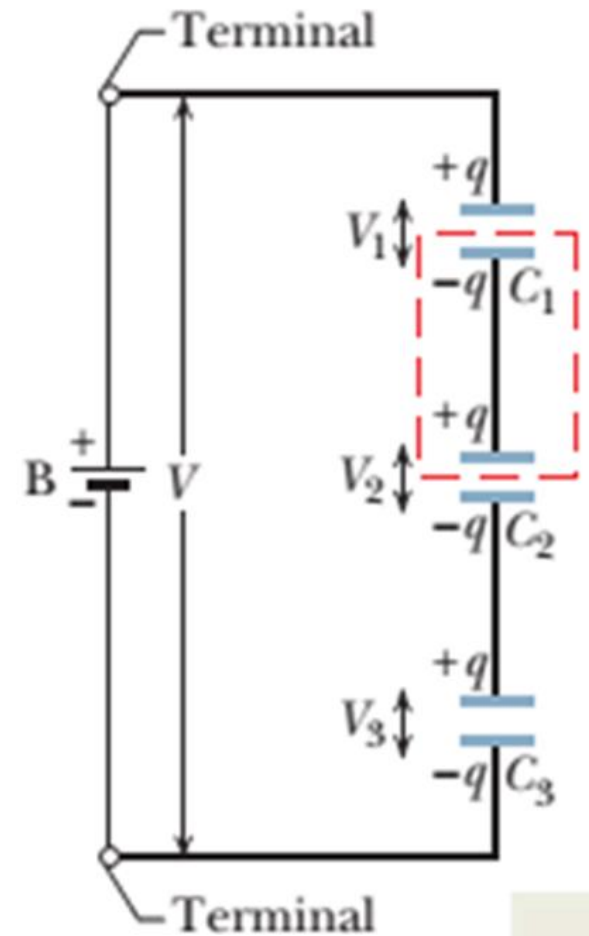
The total potential difference V due to the battery is the sum of these three potential differences. Thus,

$$V = V_1 + V_2 + V_3 = q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

The equivalent capacitance is then

$$C_{eq} = \frac{q}{V} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$



Problem 10 (Book chapter 25):

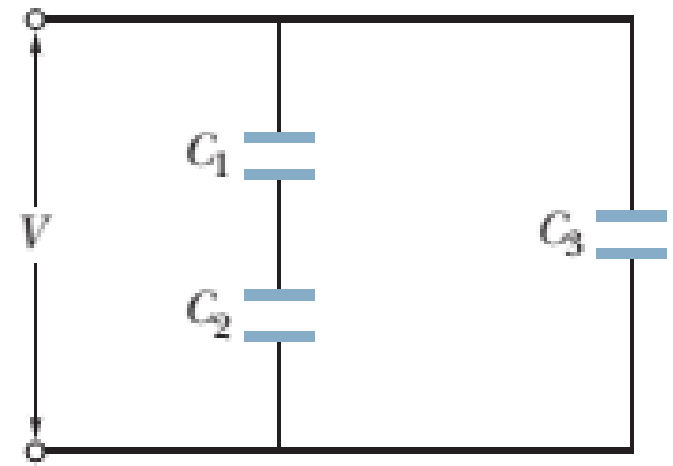
In the adjacent Figure, find the equivalent capacitance of the combination. Assume that C_1 is $10.00 \mu F$, C_2 is $5.00 \mu F$, and C_3 is $4.00 \mu F$.

Answer: **Step1:** C_1 and C_2 are in series

$$\frac{1}{C_{12}} = \frac{1}{C_1} + \frac{1}{C_2} \quad \text{or} \quad C_{12} = \frac{C_1 C_2}{C_2 + C_1}$$
$$C_{12} = \frac{(10)(5)}{5 + 10} = \frac{50}{15} = 3.33 \mu F$$

Step2: C_{12} and C_3 are in parallel

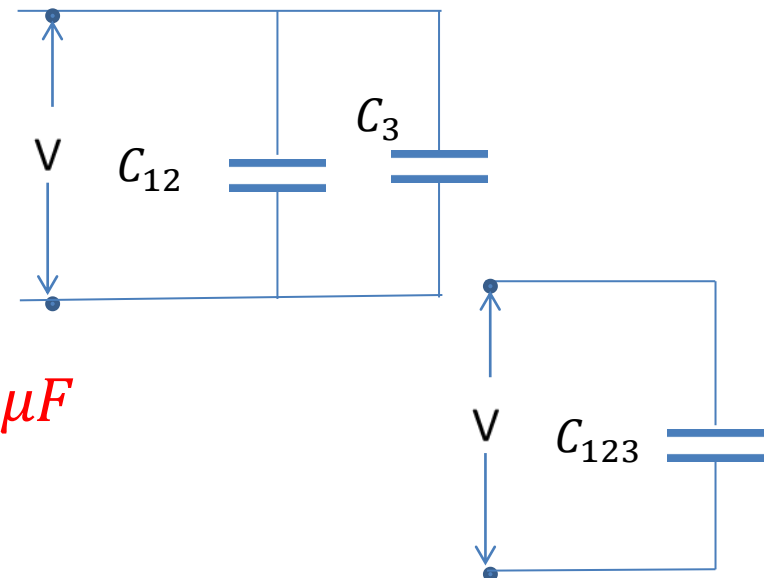
$$C_{123} = C_{12} + C_3 = 3.33 + 4 = 7.33 \mu F$$



Given

$$C_1 = 10 \mu F \quad C_2 = 5 \mu F$$

$$C_3 = 4 \mu F$$



Problem 11 (Book chapter 25):

In the adjacent Figure, find the equivalent capacitance of the combination. Assume that C_1 is $10.00 \mu F$, C_2 is $5.00 \mu F$, and C_3 is $4.00 \mu F$.

Answer:

Step1: C_1 and C_2 are in parallel

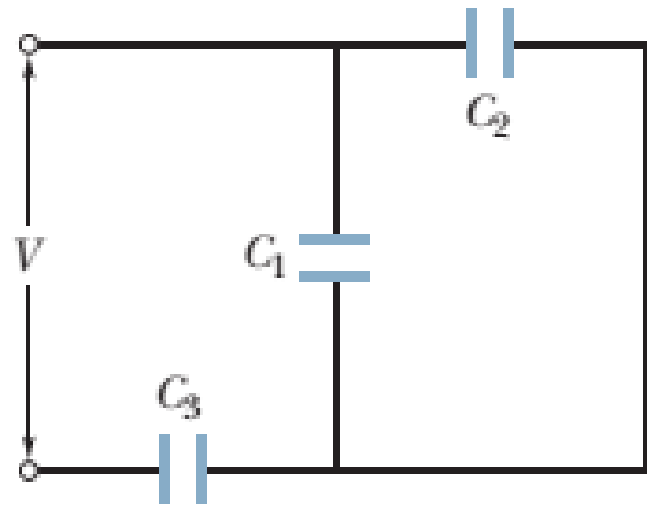
$$C_{12} = C_1 + C_2 = 10 + 5 = 15 \mu F$$

Step2: C_{12} and C_3 are in series

$$\frac{1}{C_{123}} = \frac{1}{C_{12}} + \frac{1}{C_3} = \frac{C_3 + C_{12}}{(C_{12})(C_3)}$$

$$C_{123} = \frac{(C_{12})(C_3)}{C_3 + C_{12}}$$

$$C_{123} = \frac{(15)(4)}{4 + 15} = \frac{60}{19} = 3.158 \mu F$$



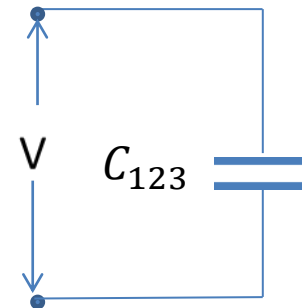
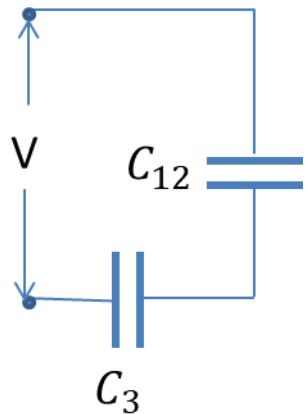
Given

$$C_1 = 10 \mu F$$

$$C_2 = 5 \mu F$$

$$C_3 = 4 \mu F$$

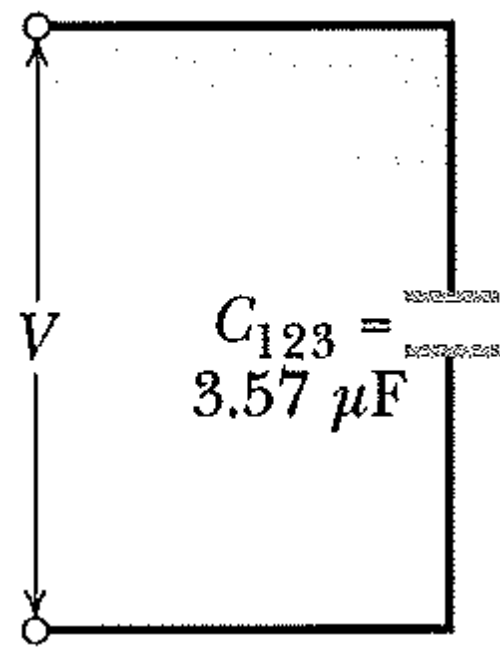
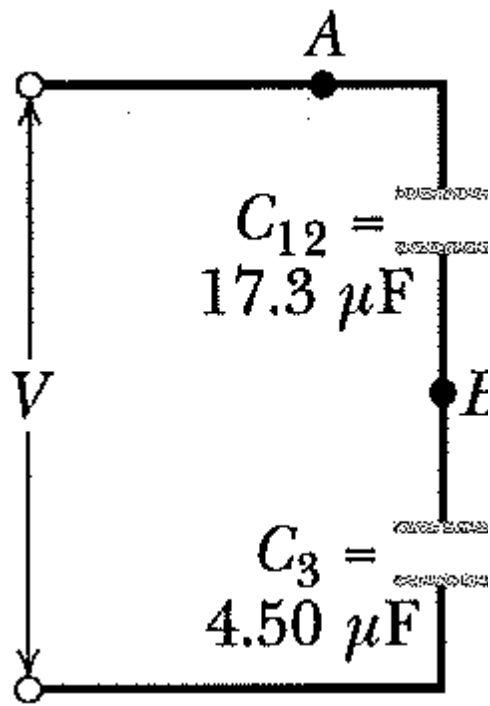
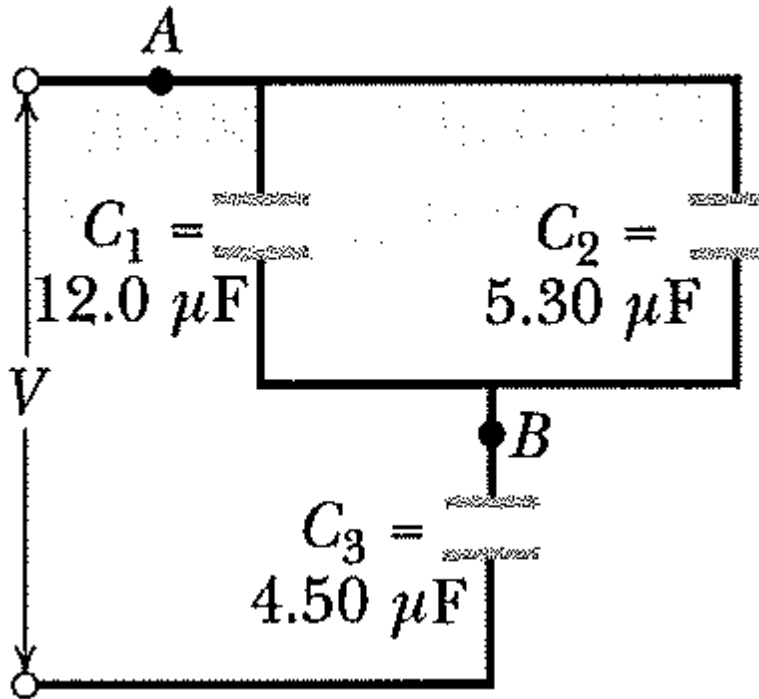
$$C_{123} = ?$$



Sample Problem 25.02(a) (page 726): Find the equivalent capacitance for the combination of capacitances shown in Figure, across which potential difference V is applied.

Assume $C_1 = 12.0 \mu F$, $C_2 = 5.30 \mu F$, and $C_3 = 4.5 \mu F$.

Home Task

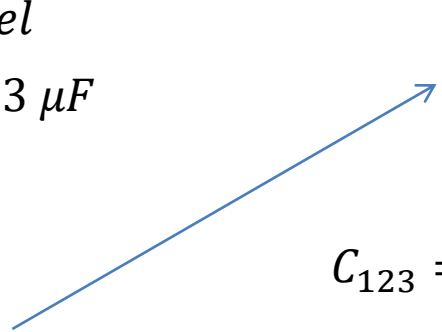


Step1: C_1 and C_2 are in parallel

$$C_{12} = C_1 + C_2 = 12 + 5.3 = 17.3 \mu F$$

Step2: C_{12} and C_3 are in series

$$\frac{1}{C_{123}} = \frac{1}{C_{12}} + \frac{1}{C_3} = \frac{C_3 + C_{12}}{(C_{12})(C_3)}$$

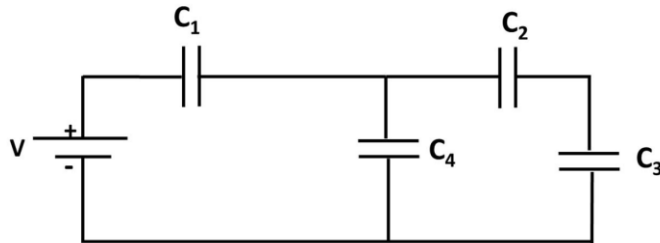


$$C_{123} = \frac{(C_{12})(C_3)}{C_3 + C_{12}}$$

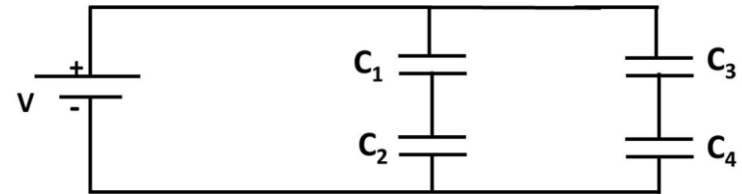
$$C_{123} = \frac{(17.3)(4.5)}{4.5 + 17.3} = \frac{77.85}{21.8} = 3.57 \mu F$$

Do by yourself....

1. Calculate the equivalent capacitance of the following two circuits.

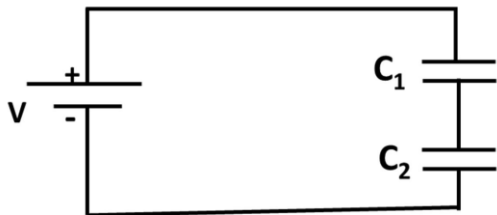


Assume that $C_1 = 4 \text{ nF}$, $C_2 = 5 \text{ nF}$, $C_3 = 9 \text{ nF}$ and $C_4 = 6 \text{ nF}$.

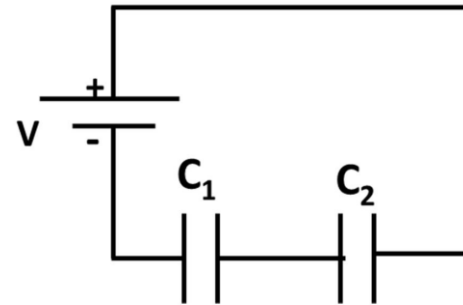


Assume that $C_1 = 2 \text{ }\mu\text{F}$, $C_2 = 4 \text{ }\mu\text{F}$, $C_3 = 6 \text{ }\mu\text{F}$ and $C_4 = 8 \text{ }\mu\text{F}$.

2. Calculate the charges on C_1 and C_2 capacitors for the following two circuits.



Assume that $C_1 = 2 \text{ }\mu\text{F}$, $C_2 = 4 \text{ }\mu\text{F}$, and $V = 8 \text{ V}$



Assume that $C_1 = 3 \text{ }\mu\text{F}$, $C_2 = 6 \text{ }\mu\text{F}$, and $V = 10 \text{ V}$

Energy Stored in an Electric Field:

Suppose that, at a given instant, a charge q' has been transferred from one plate of a capacitor to the other. The potential difference V' between the plates at that instant will be $\frac{q'}{C}$. If an extra increment of charge dq' is then transferred, the increment of work required will be,

$$dW = V' dq' = \frac{q'}{C} dq'$$

The work required to bring the total capacitor charge up to a final value q is

$$W = \int_{q'=0}^{q'=q} dW = \frac{1}{C} \int_0^q q' dq' = \frac{q^2}{2C}$$

This work is stored as potential energy U in the capacitor, so

$$U = W = \frac{q^2}{2C} = \frac{(CV)^2}{2C} = \frac{1}{2} CV^2$$

Finally,

$$U = \frac{q^2}{2C}$$

OR

$$U = \frac{1}{2} CV^2$$

Energy Density:

In a parallel-plate capacitor, the electric field has the same value at all points between the plates. Thus, **the energy density u that is, the potential energy per unit volume** between the plates should also be uniform. We can find u by dividing the total potential energy by the volume Ad of the space between the plates. Thus

$$u = \frac{U}{Ad} = \left(\frac{1}{2}CV^2\right) \frac{1}{Ad}$$

For parallel plate capacitor, the capacitance can be expressed as

$$C = \frac{\epsilon_0 A}{d}$$

Therefore,

$$u = \left(\frac{1}{2}CV^2\right) \frac{1}{Ad} = \left(\frac{V^2}{2Ad}\right) \left(\frac{\epsilon_0 A}{d}\right) = \frac{\epsilon_0 V^2}{2d^2}$$

Finally,

$$u = \frac{1}{2} \epsilon_0 \left(\frac{V}{d}\right)^2$$

OR

$$u = \frac{1}{2} \epsilon_0 E^2$$

[since $V = Ed$]

Let's try !!!

Problem 29 (Book chapter 25):

What capacitance is required to store an energy of 10 kW.h at a potential difference of 1000 V?

Problem 31 (Book chapter 25):

A 2.0 μF capacitor and a 4.0 μF capacitor are connected in parallel across a 300 V potential difference. Calculate the total energy stored in the capacitors.

Problem 32 (Book chapter 25):

A parallel-plate air-filled capacitor having area 40 cm^2 and plate spacing 1.0 mm is charged to a potential difference of 600 V. Find (a) the capacitance, (b) the magnitude of the charge on each plate, (c) the stored energy, (d) the electric field between the plates, and (e) the energy density between the plates.

Problem 33 (Book chapter 25):

A charged isolated metal sphere of diameter 10 cm has a potential of 8000 V relative to $V = 0$ at infinity. Calculate the energy density in the electric field near the surface of the sphere.

Sample problem 25.04 (page-730):

An isolated conducting sphere whose radius R is 6.85 cm has a charge $q = 1.25$ nC. (a) How much potential energy is stored in the electric field of this charged conductor? (b) What is the energy density at the surface of the sphere?

Problem 29 (Book chapter 25):

Answer:

We know

$$U = \frac{1}{2} CV^2$$

$$C = \frac{2U}{V^2} = \frac{2 \times 36 \times 10^6}{(1000)^2} = 72 \text{ F}$$

Problem 31 (Book chapter 25):

Answer:

C_1 and C_2 are in parallel

$$C_{12} = C_1 + C_2 = 2 + 4 = 6 \mu\text{F} = 6 \times 10^{-6} \text{ F}$$

We know

$$U = \frac{1}{2} C_{12} V^2 = \frac{6 \times 10^{-6} \times (300)^2}{2}$$

$$U = 0.27 \text{ J}$$

Given

$$U = 10 \text{ kW}. h = 10000 \times 60 \times 60 \text{ W} . s$$

$$U = 36 \times 10^6 \text{ W} . s$$

Potential difference, $V = 1000 \text{ volt}$

$$C = ?$$

Given

$$C_1 = 2 \mu\text{F} \quad \text{and} \quad C_2 = 4 \mu\text{F}$$

$$V = 300 \text{ volt}$$

$$U = ?$$

Problem 32 (Book chapter 25):

Given

Answer:

$$A = 40 \text{ cm}^2 = 40 \times (10^{-2})^2 \text{ m}^2 = 4 \times 10^{-3} \text{ m}^2$$

$$d = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$$

$$V = 600 \text{ volt}$$

$$C = \frac{\epsilon_0 A}{d} = \frac{8.854 \times 10^{-12} \times 4 \times 10^{-3}}{1 \times 10^{-3}}$$

$$C = 35.416 \times 10^{-12} \text{ F}$$

(a) $C = ?$

$$q = CV = 35.416 \times 10^{-12} \times 600 = 21.25 \times 10^{-9} \text{ C}$$

(b) $q = ?$

$$U = \frac{1}{2} CV^2 = \frac{35.416 \times 10^{-12} \times (600)^2}{2} = 12.75 \times 10^{-6} \text{ J}$$

(c) $U = ?$

$$E = \frac{V}{d} = \frac{600}{1 \times 10^{-3}} = 600 \times 10^3 \frac{\text{V}}{\text{m}}$$

(d) $E = ?$

$$u = \frac{1}{2} \epsilon_0 \left(\frac{V}{d} \right)^2 = \frac{8.854 \times 10^{-12} \times (600)^2}{2 \times (10^{-3})^2} = 1.59 \frac{\text{J}}{\text{m}^3}$$

(e) $u = ?$

Problem 33 (Book chapter 25):

Answer:

$$u = \frac{1}{2} \epsilon_0 \left(\frac{V}{R} \right)^2 = \frac{8.854 \times 10^{-12} \times (8000)^2}{2 \times (0.05)^2}$$

$$u = \frac{566.66 \times 10^{-6}}{5 \times 10^{-3}} = 113.33 \times 10^{-3} \frac{J}{m^3}$$

Sample problem 25.04 (page-730):

Answer:

We know

$$U = \frac{q^2}{2C}$$

For isolated sphere, $C = 4\pi\epsilon_0 R$

$$U = \frac{q^2}{2 \times 4\pi\epsilon_0 R} = \frac{9 \times 10^9 \times (1.25 \times 10^{-9})^2}{2 \times 0.0685}$$

$$U = 102.64 \times 10^{-9} J$$

$$u = \frac{1}{2} \epsilon_0 E^2 = \frac{8.854 \times 10^{-12}}{2} \left(\frac{q}{4\pi\epsilon_0 R^2} \right)^2 = \frac{8.854 \times 10^{-12}}{2} \left(\frac{9 \times 10^9 \times 1.25 \times 10^{-9}}{(0.0685)^2} \right)^2$$

$$u = \frac{1120.58 \times 10^{-12}}{4.403 \times 10^{-5}} = 25.450 \times 10^{-6} \frac{J}{m^3}$$

Given

For isolated sphere:

Diameter, $D = 10 \text{ cm}$

Radius, $R = 5 \text{ cm} = 0.05 \text{ m}$

$V = 8000 \text{ volt}$

$u = ?$

Given

For isolated sphere:

Radius, $R = 6.85 \text{ cm} = 0.0685 \text{ m}$

$q = 1.25 \text{ nC} = 1.25 \times 10^{-9} \text{ C}$

(a) $U = ?$

(b) $u = ?$



Thank You